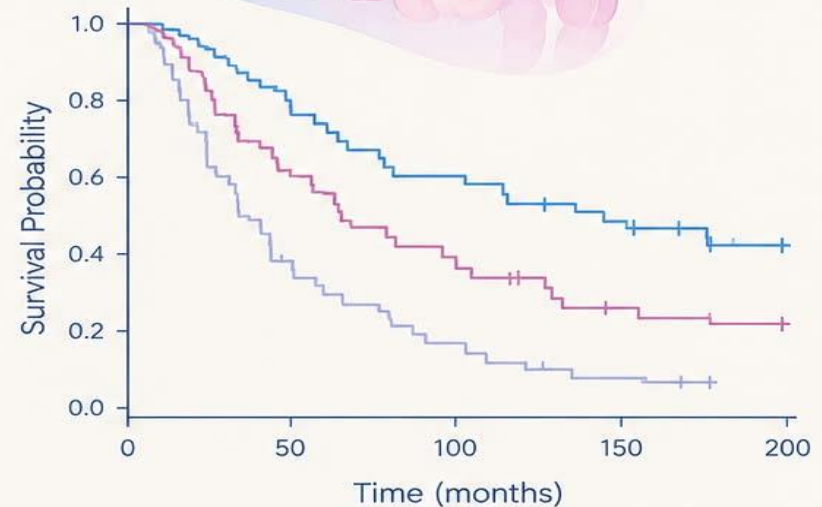
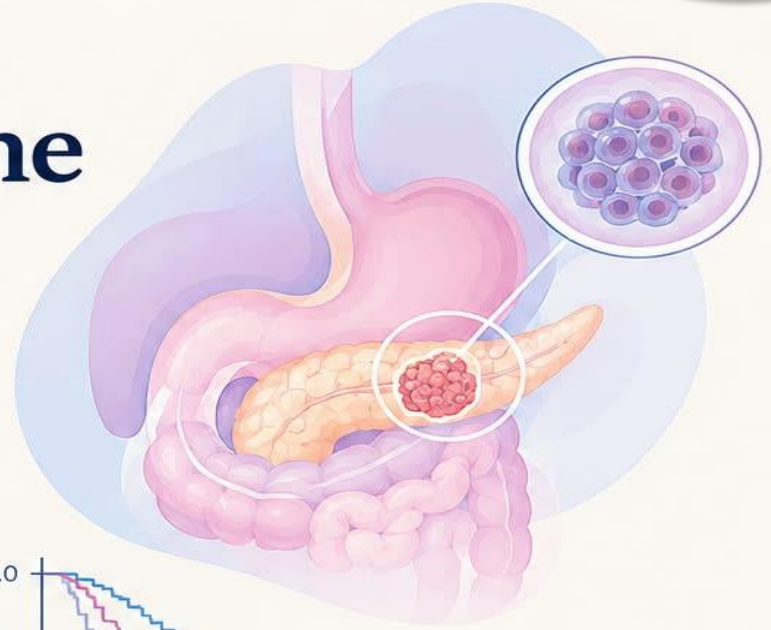


SEER Neuroendocrine Tumor

Prognostic Factor Survival Analysis

Exploring clinical and demographic factors associated with survival in neuroendocrine tumor patients using SEER data.



SEER
Data



Survival
Analysis



R



Neuroendocrine
Tumors

142,420 patient record
2000–2023

Swipe →

01 • BACKGROUND

Why study neuroendocrine tumours?

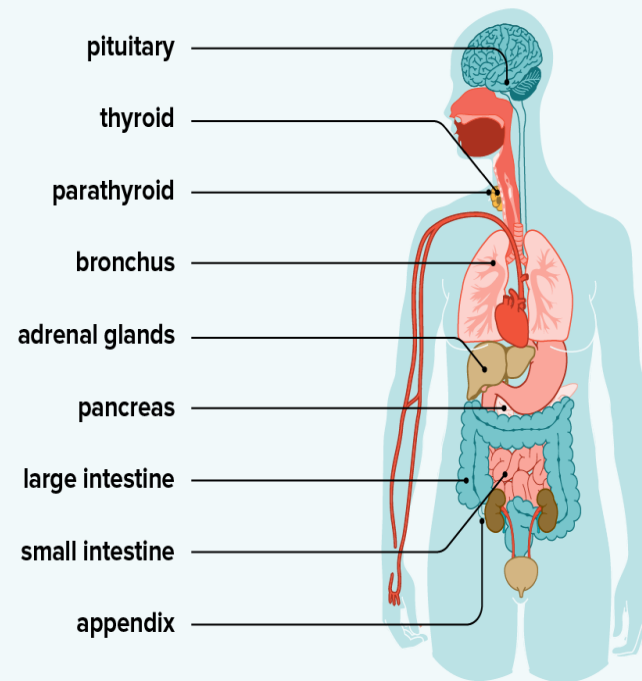
A rare and biologically diverse cancer group.

THE BIOLOGY

NETs arise from cells with both nerve-like and hormone-producing characteristics.

- Can develop in multiple organs
- Biologically heterogeneous
- Prognosis varies substantially
- Population-level evidence can reveal broad survival patterns

Most common areas where neuroendocrine tumors appear



healthline

02 • STUDY AIM

Research questions & objectives

The analysis connects tumour biology with time-to-event statistics.

RESEARCH QUESTION 01

Do tumour site and stage at diagnosis predict overall survival in NET patients?

RESEARCH QUESTION 02

Does patient age confound the relationship between stage and survival?

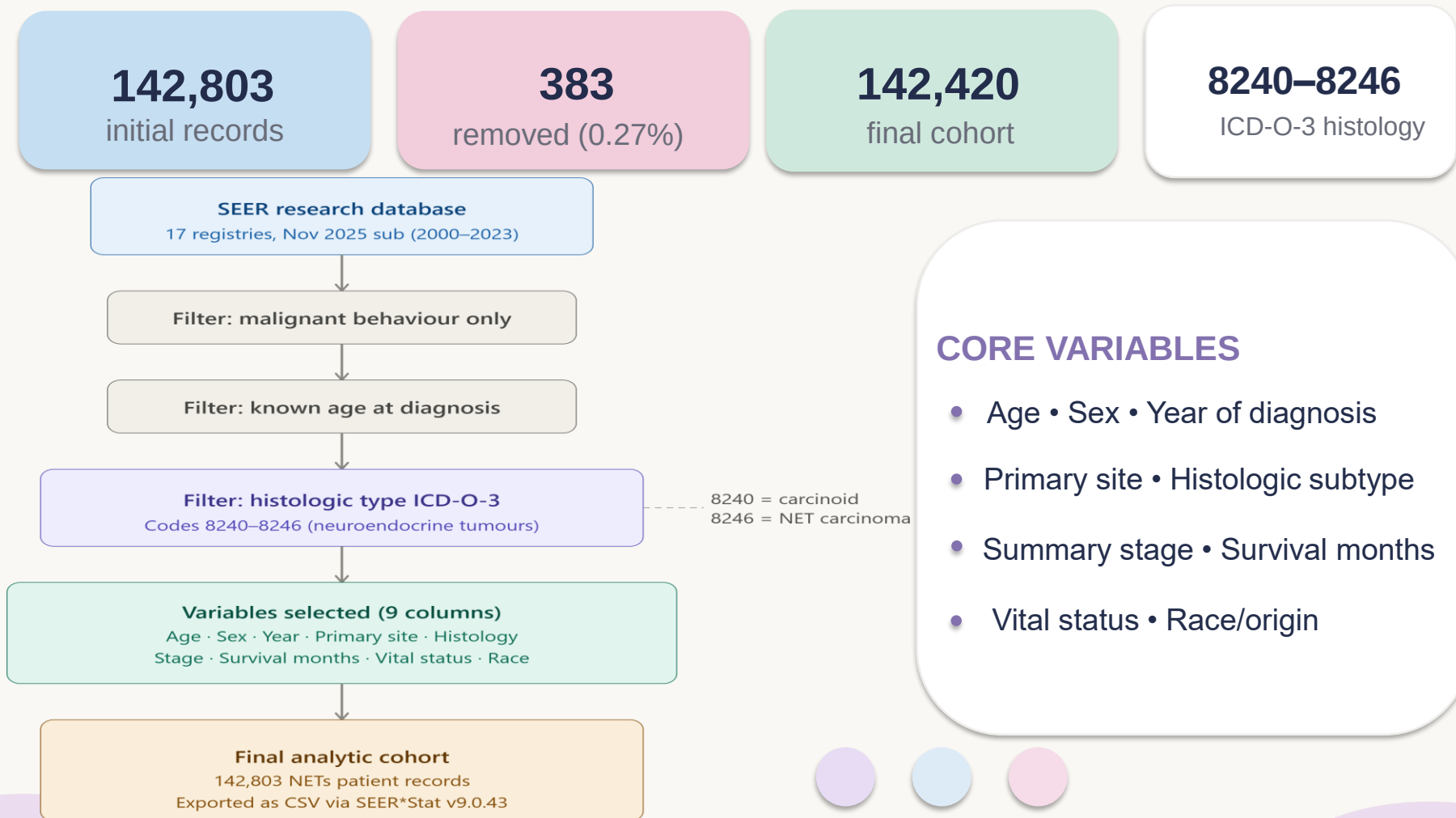
ADDITIONAL OBJECTIVES

- Evaluate age, sex, race and histologic subtype
- Compare crude and adjusted associations
- Examine tumour site × stage patterns
- Demonstrate why censoring-aware survival methods matter
- Assess the proportional hazards assumption

03 • DATASET

SEER data & study cohort

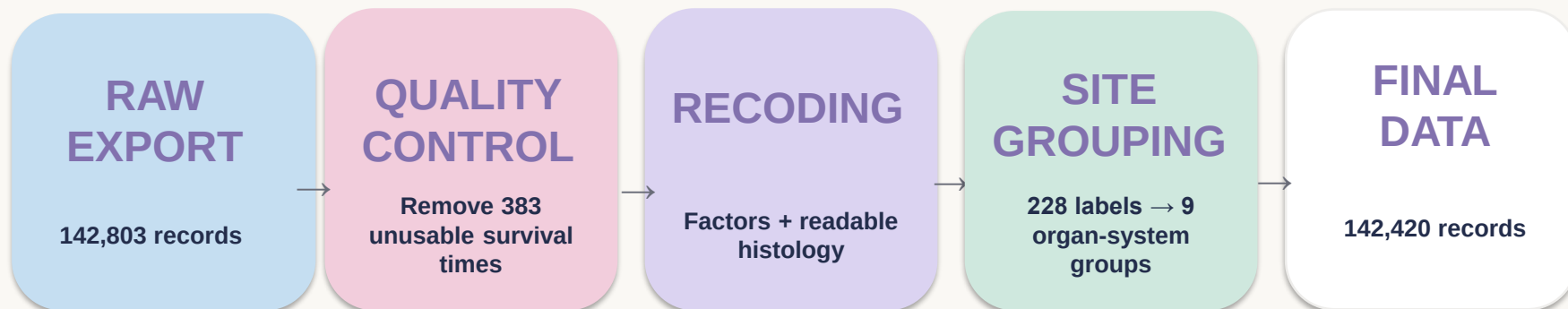
Population-based U.S. cancer registry data, 2000–2023.



04 • PHASE 1

Data cleaning & preparation

Turning the raw SEER export into an analysis-ready dataset.



KEY CLEANING DECISIONS

Ultra-rare histologies (n=62 and n=8) were collapsed into “Other rare NET subtype”; categorical variables were converted to factors; survival time and event status were prepared for time-to-event analysis.

```

> glimpse(df)
Rows: 142,420
Columns: 14
 $ age_recode_with_1_year_old_and_90
 $ sex
 $ year_of_diagnosis
 $ primary_site_labeled
 $ histologic_type_icd_o_3
 $ summary_stage_2000_1998_2017
 $ survival_months
 $ vital_status_recode_study_cutoff_used
 $ race_and_origin_recode_nhw_nhb_nhaian_nhapi_hispanic
 $ death_event
 $ site_group
 $ age_group
 $ age_group_v2
 $ histology_factor

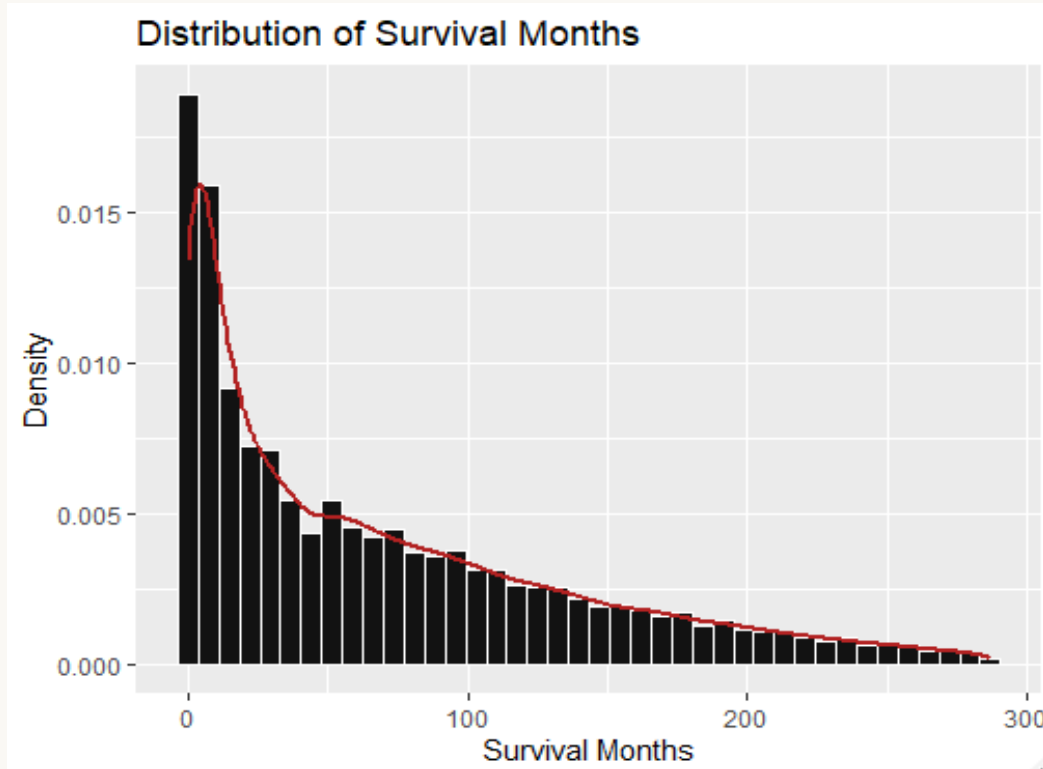
```

<ord> 65-69 years, 65-69 years, 70-74 years, 80...
 <fct> Female, Male, Female, Female, Female, Mal...
 <int> 2008, 2011, 2004, 2010, 2007, 2023, 2005,...
 <chr> "C20.9-Rectum, NOS", "C17.9-Small intesti...
 <fct> "Neuroendocrine carcinoma, NOS", "Neuroen...
 <fct> Localized, Distant, Distant, Localized, D...
 <dbl> 120, 144, 67, 68, 17, 3, 0, 170, 15, 130,...
 <fct> Dead, Alive, Dead, Dead, Dead, Alive, Dea...
 <fct> Non-Hispanic Asian or Pacific Islander, N...
 <dbl> 1, 0, 1, 1, 1, 0, 1, 0, 1, 1, 1, 1, 1, 0,...
 <fct> Rectum, Small intestine, Lung/Bronchus, L...
 <fct> 65-69 years, 65-69 years, 70-74 years, 80...
 <fct> 65-69 years, 65-69 years, 70-74 years, 80...
 <fct> "Neuroendocrine carcinoma, NOS", "Neuroen...

05 • PHASE 2

Exploratory data analysis

Understanding the cohort before fitting survival models.



Histogram + density: Survival months

WHAT WAS CHECKED?

- Descriptive statistics & frequencies
- Sex × Stage
- Site × Stage
- Race × Stage
- ANOVA / Pearson correlation for diagnosis year

Goal: identify structure, imbalance and potential confounding before modelling.

06 • PHASE 3

Kaplan–Meier survival analysis

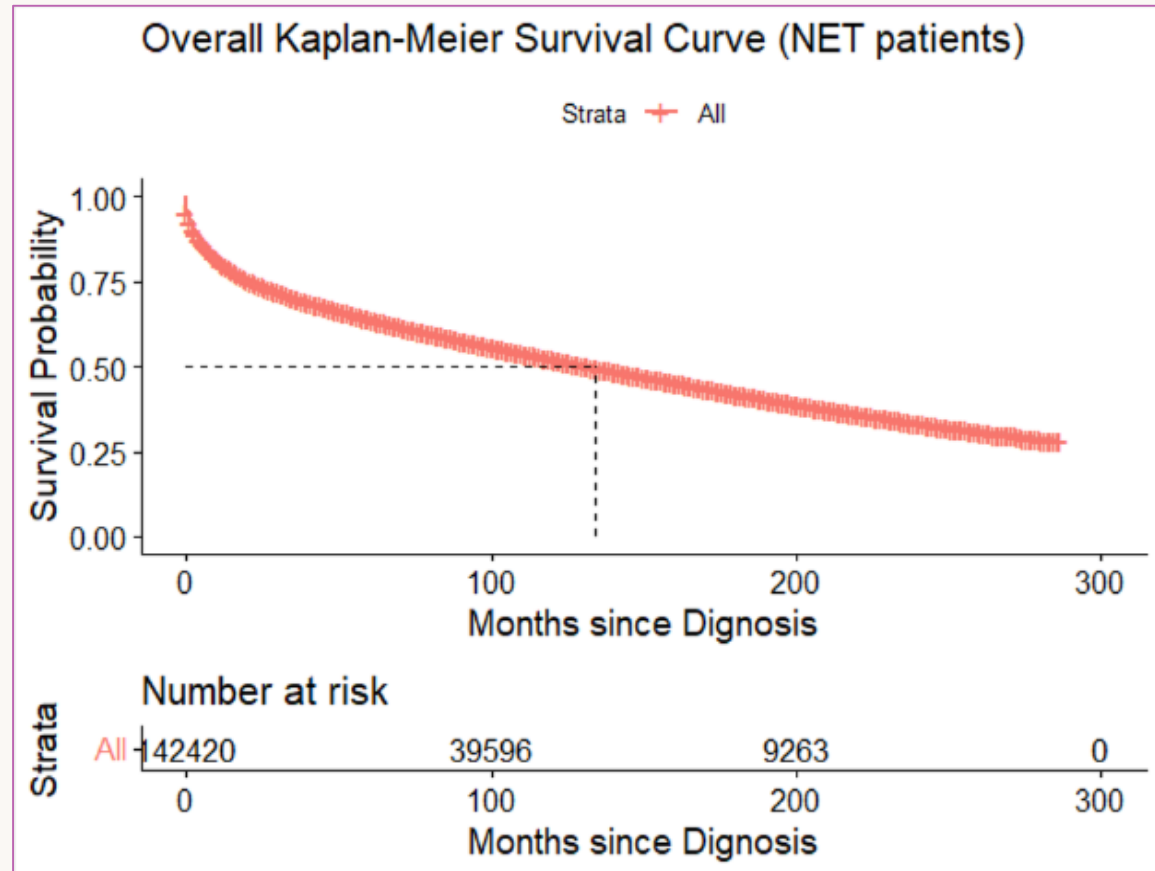
Estimating overall survival while retaining follow-up time and censoring.

SURVIVAL OBJECT**Time****+****Death event /
censoring**

Kaplan–Meier estimates the probability of surviving beyond each time point.

**GROUP
COMPARISONS****Stage
Site group**

Log-rank tests assess whether survival curves differ between groups.



07 • PHASE 3 RESULTS

Stage is strongly associated with survival

The largest survival separation in the project.

LOG-RANK TEST

$\chi^2 = 31,505$

$p < 0.0001$

~20× difference in median survival between localized and distant disease.

242

months • Localized

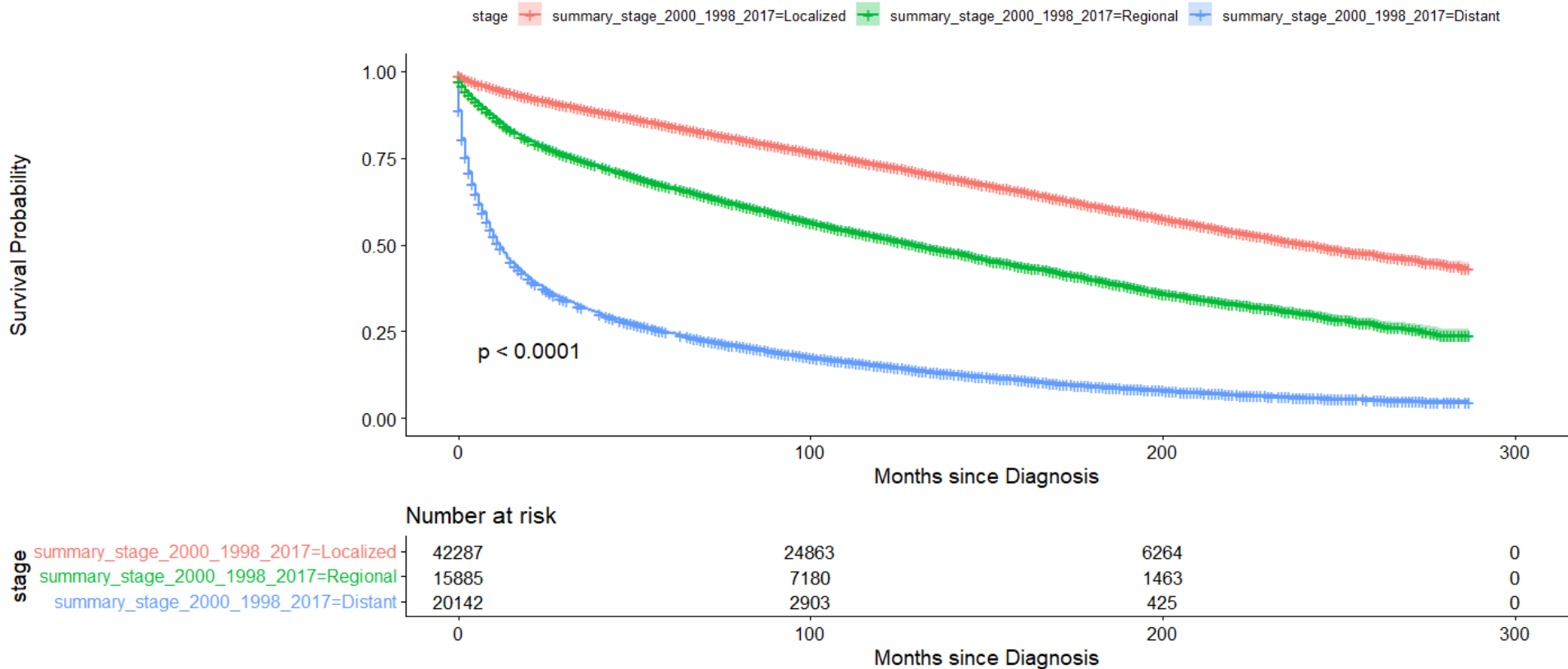
130

months • Regional

12

months • Distant

Kaplan-Meier Survival by Stage (Localized/Regional/Distant)



08 • PHASE 3 RESULTS

Tumour site and survival

Survival also varied across the nine clinically meaningful site groups.

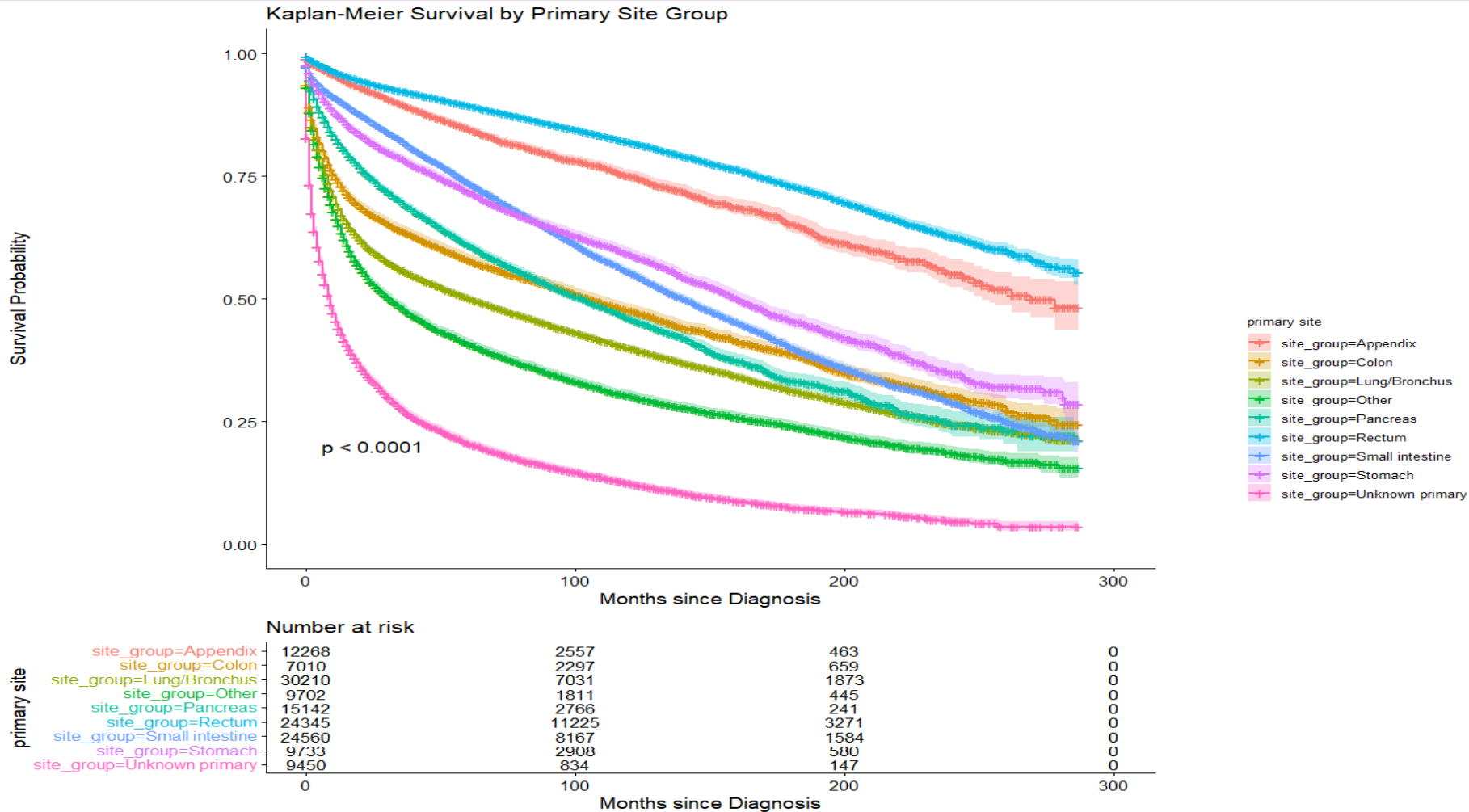
KEY RESULT

Unknown primary site was the single worst prognostic group.

Median survival: 9 months

Univariate HR: 9.08 vs. Appendix

Its prognosis is clinically consistent with advanced, widely spread disease.



09 • PHASE 4

Cox proportional hazards regression

Moving from group comparisons to adjusted prognostic effects.

MODEL STRATEGY

Univariate screening



Multivariable model

Stage + Site + Age + Sex + Race +
Histology

Age + Race + Stage + Site + Histology + Sex →

Cox model →

Hazard Ratios →

Prognostic factors

WHAT DOES HR MEAN?

Hazard Ratio (HR)

Relative instantaneous
risk of the event,
comparing one group with
a reference group.

HR > 1 → higher hazard

HR < 1 → lower hazard

```
cox_multi <- coxph(Surv(survival_months, death_event) ~  
  summary_stage_2000_1998_2017 +  
  site_group +  
  age_group_v2 +  
  sex +  
  race_and_origin_recode_nhw_nhb_nhaian_nhapi_hispanic +  
  factor(histologic_type_icd_o_3),  
  data = df_known_stage)  
summary(cox_multi)
```

10 • PHASE 4 RESULTS

Adjusted prognostic factors

Stage and site remained the dominant predictors after adjustment.

MAIN FINDINGS

Distant stage

HR 4.33

95% CI 4.21–4.44

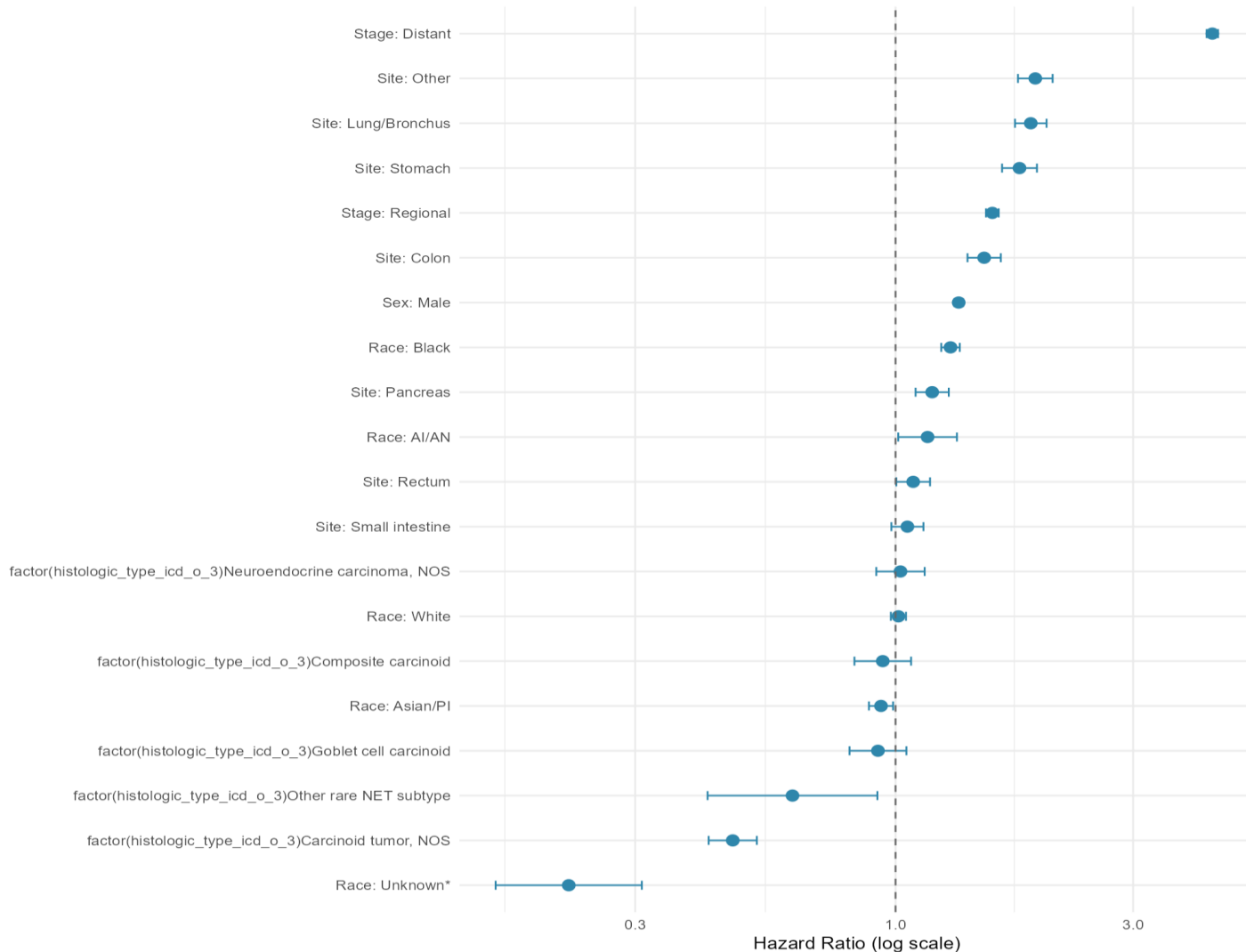
Lung/Bronchus & Other sites

≈1.9× hazard vs. Appendix

Several crude associations shrank toward the null after adjustment.

Adjusted Hazard Ratios (Final Cox Model)

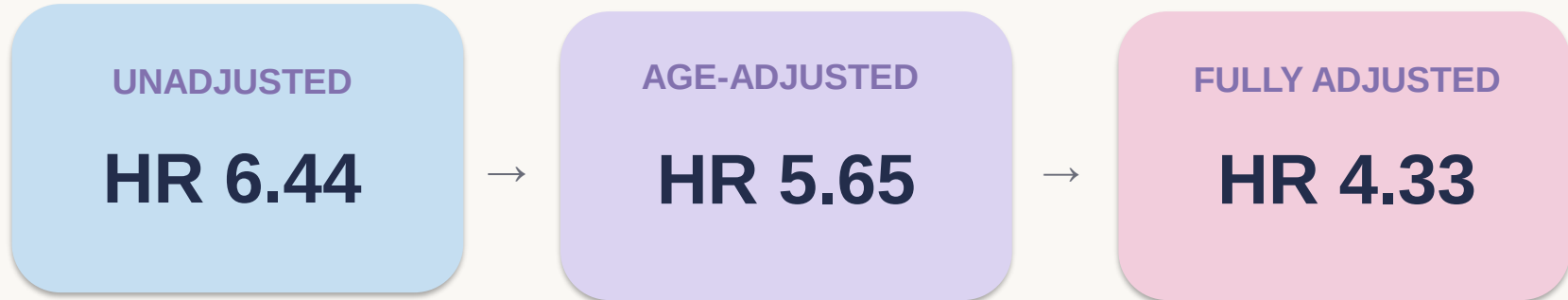
*Unknown Race treated with caution — likely a data-quality artifact



11 • PHASE 4 RESULTS

Confounding: the role of age

Adjusting for age reduced—but did not explain—the stage effect.



INTERPRETATION

Age explained part of the crude stage–survival association. The remaining reduction reflects the combined influence of site, sex, race and histologic subtype.

```
#Step 3 – Isolating Age's confounding effect on Stage specifically
cox_stage_only <- coxph(Surv(survival_months, death_event) ~ summary_stage_2000_1998_2017,
                        data = df_known_stage)
cox_stage_age <- coxph(Surv(survival_months, death_event) ~ summary_stage_2000_1998_2017 + age_group_v2,
                      data = df_known_stage)

summary(cox_stage_only)$coefficients
summary(cox_stage_age)$coefficients
```

12 • PHASE 4 RESULTS

Why censoring matters

A direct comparison using the project's own data.

COX REGRESSION

Uses time + event status

Handles censoring

Produces hazard ratios

NAIVE LOGISTIC REGRESSION

Uses dead/alive only

Ignores follow-up time

Produces odds ratios

Age 90+ • Cox HR = 8.23 vs. Logistic OR = 90.45

13 • PHASE 5

Model diagnostics

Checking whether the Cox model's assumptions are reasonable.

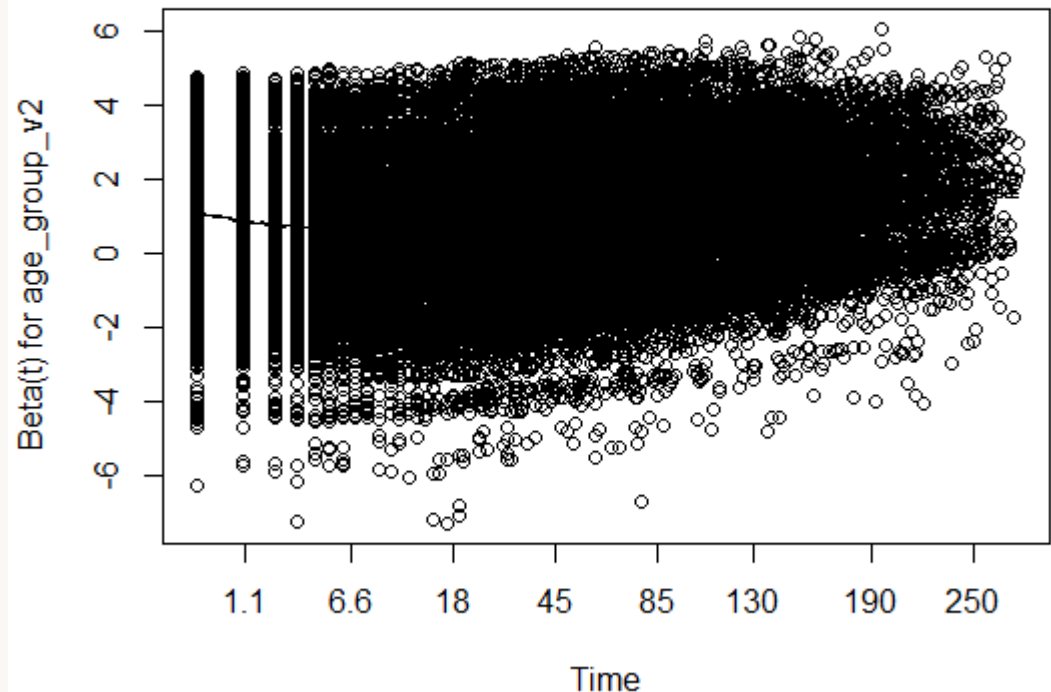
PROPORTIONAL HAZARDS

Schoenfeld residuals

Tests whether predictor effects remain approximately constant over time.

Age showed a clear violation.

→ stratified in final model



14 • PHASE 5 INTERPRETATION

Diagnostics & model trade-offs

Large samples make statistical assumption tests especially sensitive.

AGE

Clear time-dependence

Handled by stratification in the final model.

Trade-off:

No explicit age HR
Concordance: 0.812 → 0.759

STAGE • SITE • HISTOLOGY

Statistically significant departures

Diagnostic plots did not show equally strong visual evidence of major non-proportionality.

Interpreted as average effects over follow-up.

MODEL CONTEXT

- Large N increases sensitivity of tests
- Not every statistical departure implies severe practical bias
- Assumptions and limitations are reported transparently

15 • PHASE 6

Final visualization & Table 1

Turning the statistical output into publication-style reporting.

Characteristic	Overall N = 78,314 ¹	Localized N = 42,287 ¹	Regional N = 15,885 ¹	Distant N = 20,142 ¹	p-value ²
Sex					<0.001
Female	41,352 (53%)	23,535 (56%)	8,071 (51%)	9,746 (48%)	
Male	36,962 (47%)	18,752 (44%)	7,814 (49%)	10,396 (52%)	
Age group					<0.001
00-09 years	31 (<0.1%)	27 (<0.1%)	3 (<0.1%)	1 (<0.1%)	
10-14 years	198 (0.3%)	161 (0.4%)	30 (0.2%)	7 (<0.1%)	
15-19 years	439 (0.6%)	352 (0.8%)	63 (0.4%)	24 (0.1%)	
20-24 years	662 (0.8%)	531 (1.3%)	92 (0.6%)	39 (0.2%)	
25-29 years	1,069 (1.4%)	785 (1.9%)	184 (1.2%)	100 (0.5%)	
30-34 years	1,469 (1.9%)	1,005 (2.4%)	312 (2.0%)	152 (0.8%)	
35-39 years	2,209 (2.8%)	1,473 (3.5%)	407 (2.6%)	329 (1.6%)	
40-44 years	3,416 (4.4%)	2,137 (5.1%)	656 (4.1%)	623 (3.1%)	
45-49 years	5,265 (6.7%)	3,108 (7.3%)	1,033 (6.5%)	1,124 (5.6%)	
50-54 years	10,051 (13%)	6,579 (16%)	1,695 (11%)	1,777 (8.8%)	
55-59 years	9,826 (13%)	5,529 (13%)	1,917 (12%)	2,380 (12%)	
60-64 years	10,174 (13%)	5,271 (12%)	2,125 (13%)	2,778 (14%)	
65-69 years	10,511 (13%)	5,269 (12%)	2,201 (14%)	3,041 (15%)	

16 • ADDITIONAL FINDING

Tumour site shapes stage at diagnosis

A striking site–stage relationship helps explain survival differences.

93.4%

Rectal NETs localized at diagnosis

38.9%

Pancreatic NETs localized at diagnosis

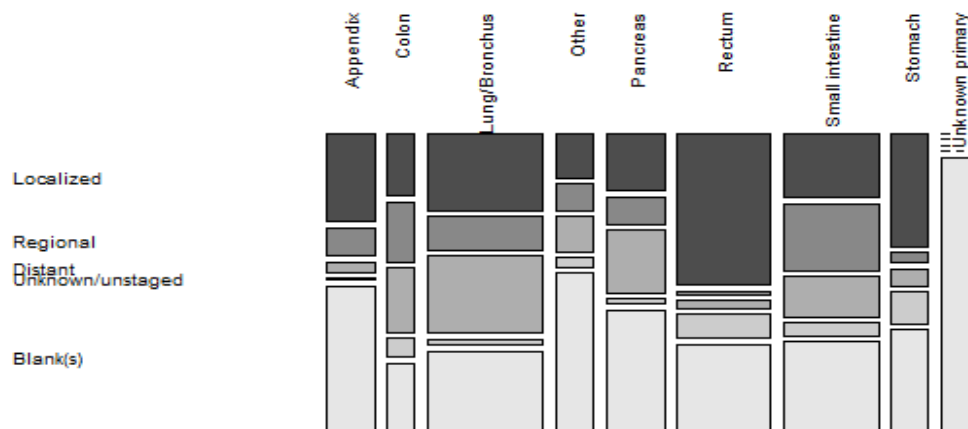
Mosaic plot & heatmap

INTERPRETATION

Differences in symptom visibility, screening, incidental detection and presentation may contribute to different stage distributions across organs.

Key insight: tumour site may influence survival partly through its relationship with stage.

Site Group vs Summary Stage



17 • LIMITATIONS

What the data cannot tell us

Transparent interpretation is essential for observational registry data.

MISSING STAGE

~45% had blank/unknown stage

OBSERVATIONAL DESIGN

Associations ≠ causation

DATA-QUALITY ARTIFACTS

Unknown race / unknown primary
require caution

FOLLOW-UP VARIES

Later diagnosis years have less
possible follow-up

Race-based findings were treated cautiously because socioeconomic and healthcare-access factors are not fully captured in SEER.

18 • KEY FINDINGS

What did the analysis show?

Five findings connect the clinical story to the statistical evidence.

01

Stage

Distant disease had a median survival of 12 months vs. 242 months for localized disease.

02

Site

Tumour site was strongly associated with stage and survival.

03

Adjustment

Stage and site remained dominant after multivariable adjustment.

04

Confounding

Age explained part, but not all, of the crude stage effect.

05

Censoring

Ignoring follow-up time produced substantially more extreme estimates.



CONCLUSION

Stage & tumour site are central to NET survival patterns

A reproducible R workflow using SEER data demonstrated how Kaplan–Meier curves, Cox regression and model diagnostics can uncover clinically meaningful population-level survival patterns.

Time-to-event analysis preserves information that simple dead/alive models discard.



Thank you

